Residual Analysis

1 Model Adequacy Checking

Assumptions Behind the Fitted Model

The linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is fitted under the following assumptions:

- The relationship between y and the regressors is (approximately) linear
- $\mathrm{E}[\varepsilon_i] = 0$
- $\mathrm{Var}(\varepsilon_i) = \sigma^2$ for all i (constant variance)
- The errors are uncorrelated (independence)
- The errors are normally distributed (needed for the inference)

Least squares returns estimates whether or not these assumptions hold.

What Can We Observe?

The errors ε_i are never observed. What can we observe instead?

2 Raw Residuals

Raw Residuals

The ith residual is $e_i = y_i - \hat{y}_i$. In matrix form

$$\begin{aligned}\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} &= (\mathbf{I} - \mathbf{H})\mathbf{y} \\ &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon},\end{aligned}$$

where the last step uses $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$.

Leverage

The hat matrix is $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ with diagonal elements

$$h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i.$$

- Intuitively, h_{ii} measures how far $\mathbf{x}_i$ is from the centroid of the regressor values
- h_{ii} is large when $\mathbf{x}_i$ is far from the centroid of the regressor values
- Points with unusual $\mathbf{x}_i$-values have more "leverage" to pull the fitted surface toward their own y_i
- $0 \le h_{ii} \le 1$ and $\sum_{i=1}^{n} h_{ii} = p$, where p is the number of model parameters
- $\hat{y}_i = \sum_j h_{ij} y_j$, so h_{ii} is the weight of y_i in its own fitted value

Flagging High-Leverage Points

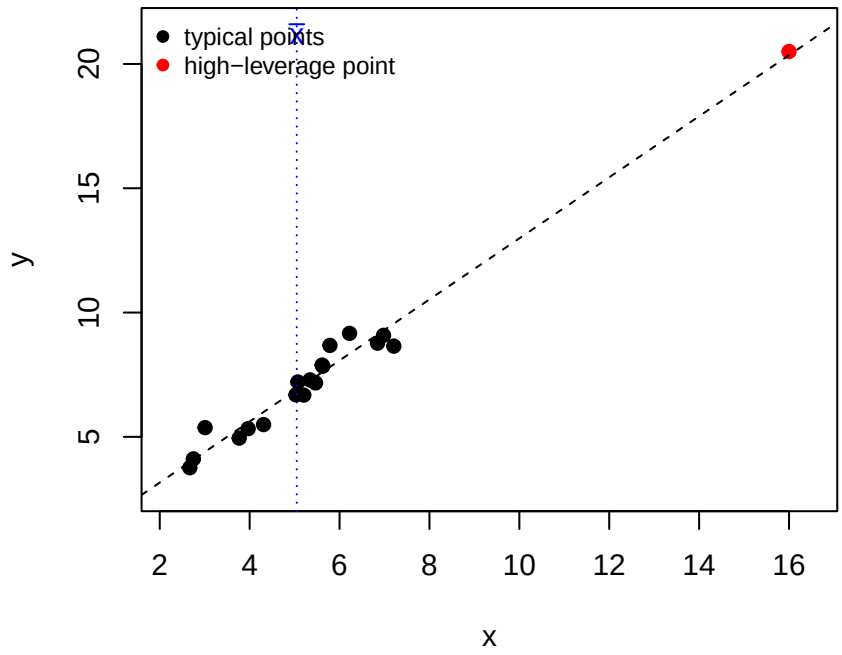
Rule of thumb: observation i has high leverage if

$$h_{ii} > \frac{2p}{n}.$$

- Delivery time data: $p = 3$, $n = 25$, so the cutoff is $2p/n = 2(3)/25 = 0.24$
- Observation 9: $h_{99} = 0.498 > 0.24$, flagged as high leverage
- High leverage is about an unusual $\mathbf{x}_i$, not an unusual y_i: a flagged point can still have a small residual if y_i happens to agree with the trend at that $\mathbf{x}_i$

Leverage: Geometric Intuition

- Leverage depends only on the location of $\mathbf{x}_i$, not on y_i
- A point far from $\bar{x}$ has high leverage, whatever its y-value
- Here the flagged point sits far from the rest of the data in x, yet close to the fitted line: high leverage, small residual



Mean and Covariance of the Residuals

$$\begin{aligned} \mathrm{E}[\mathbf{e}] &= \mathbf{0} \\ \mathrm{Var}(\mathbf{e}) &= \sigma^2(\mathbf{I} - \mathbf{H}) \end{aligned} \tag{1}$$

Element by element

$$\begin{aligned} \mathrm{Var}(e_i) &= \sigma^2(1 - h_{ii}) \\ \mathrm{Cov}(e_i, e_j) &= -\sigma^2 h_{ij}, \qquad i \neq j. \end{aligned}$$

Are the Residuals iid?

The errors are assumed iid $N(0, \sigma^2)$. Are the residuals $e_1, \ldots, e_n$ iid?

No. $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$ differs across i whenever the h_{ii} differ, $\mathrm{Cov}(e_i, e_j) = -\sigma^2 h_{ij} \neq 0$, and the residuals satisfy the p linear constraints $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$ (in particular $\sum_i e_i = 0$ when the model has an intercept).

3 Different Type of Residuals

Standardized Residuals

$$d_i = \frac{e_i}{\sqrt{\mathrm{MSE}}}, \qquad \mathrm{MSE} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n - p} \tag{2}$$

- $\mathrm{E}[d_i] = 0$ and $\mathrm{Var}(d_i) \approx 1$, and d_i is free of the units of y
- Rule of thumb: $|d_i| > 3$ marks a potential outlier
- d_i ignores that $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$ differs across observations, see (1)

Studentized Residuals

$$r_i = \frac{e_i}{\sqrt{\mathrm{MSE}\,(1 - h_{ii})}} = \frac{d_i}{\sqrt{1 - h_{ii}}} \tag{3}$$

- Divides e_i by its own estimated standard deviation from (1), so $\mathrm{Var}(r_i) = 1$ for every i, whatever the location of $\mathbf{x}_i$
- $|r_i| \geq |d_i|$, with d_i from (2)
- $r_i \approx d_i$ when h_{ii} is small, and they differ noticeably for high-leverage points

PRESS Residual: Leave-One-Out Prediction Error

The PRESS residual is the error when y_i is predicted from the model fitted to the other $n - 1$ observations

$$e_{(i)} = y_i - \widehat{y}_{(i)} = \frac{e_i}{1 - h_{ii}}.$$

- The identity avoids refitting the model n times
- A large h_{ii} inflates the gap between $e_{(i)}$ and e_i

Variance of PRESS Residuals and the PRESS Statistic

$$\mathrm{Var}(e_{(i)}) = \frac{\sigma^2}{1 - h_{ii}}$$

$$\mathrm{PRESS} = \sum_{i=1}^{n} e_{(i)}^2$$

- PRESS measures predictive ability: smaller is better

- Standardizing $e_{(i)}$ by its standard deviation returns the studentized residual

$$\frac{e_{(i)}}{\sqrt{\text{Var}(e_{(i)})}} = \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}}$$

Removing Observation i From the Variance Estimate

MSE uses all n observations, including observation i: a large e_i inflates MSE and shrinks r_i. Instead use the variance estimate from the fit without observation i

$$S^2_{(i)} = \frac{(n-p)\,\text{MSE} - e_i^2/(1-h_{ii})}{n-p-1}, \tag{4}$$

which also needs no refit.

R-Student Residuals (cont.)

$$t_i = \frac{e_i}{\sqrt{S^2_{(i)}\,(1-h_{ii})}}$$

- Replaces MSE in (3) by $S^2_{(i)}$ from (4)
- If the errors are normal, $t_i \sim t_{n-p-1}$
- More sensitive to an outlier than r_i: the outlier no longer inflates its own denominator
- Used to test for outliers (later lecture)

Types of Residuals

Residual	Formula	Variance / distribution	Note
Raw	e_i	$\sigma^2(1-h_{ii})$	Unequal variances
Standardized	$e_i/\sqrt{\text{MSE}}$	≈ 1	Ignores h_{ii}
Studentized	$e_i/\sqrt{\text{MSE}(1-h_{ii})}$	1	Accounts for leverage
PRESS	$e_i/(1-h_{ii})$	$\sigma^2/(1-h_{ii})$	Leave-one-out error
R-student	$e_i/\sqrt{S^2_{(i)}(1-h_{ii})}$	t_{n-p-1}	Outlier detection

Leverage and Residual Size

Points with large leverage tend to have small raw residuals. Why?

A large h_{ii} pulls the fitted surface toward y_i, since $\hat{y}_i = \sum_j h_{ij}y_j$ contains $h_{ii}y_i$. Formally $\text{Var}(e_i) = \sigma^2(1-h_{ii}) \to 0$ as $h_{ii} \to 1$, so e_i understates how far y_i is from where the rest of the data would put it.

4 Example: Delivery Time Data

Delivery Time Data

- y: delivery time (minutes) of a route driver servicing vending machines

- x_1: number of cases of product stocked
- x_2: distance walked by the driver (feet)
- $n = 25$ observations, $p = 3$

Fitted model

$$\hat{y} = 2.341 + 1.616\,x_1 + 0.0144\,x_2, \qquad \text{MSE} = 10.624, \quad \hat{\sigma} = 3.259.$$

Delivery Time Data: Residuals

i	h_{ii}	e_i	d_i	r_i	$e_{(i)}$	t_i
1	0.102	−5.028	−1.543	−1.628	−5.598	−1.696
9	0.498	7.420	2.276	3.214	14.789	4.311
10	0.196	2.376	0.729	0.813	2.957	0.807
15	0.041	0.671	0.206	0.210	0.700	0.206
20	0.102	−5.788	−1.776	−1.874	−6.443	−1.997
22	0.392	−3.687	−1.131	−1.450	−6.059	−1.490

Six of the 25 observations. d_i standardized, r_i studentized, $e_{(i)}$ PRESS, t_i R-student. PRESS = 459.04.

Observation 9

Observation 9 (30 cases, 1460 ft) has the largest leverage, $h_{99} = 0.498$. Its residuals are $d_9 = 2.276$, $r_9 = 3.214$, $t_9 = 4.311$. Why does the residual keep growing from d_9 to r_9 to t_9?

$r_9 = d_9/\sqrt{1 - h_{99}} = d_9/0.708$: high leverage shrinks $\text{Var}(e_9)$, so the raw residual understates how unusual the point is. t_9 replaces MSE = 10.624 by $S^2_{(9)} = 5.905$, the variance estimate without observation 9 itself: the outlier no longer inflates the denominator.

5 Normality

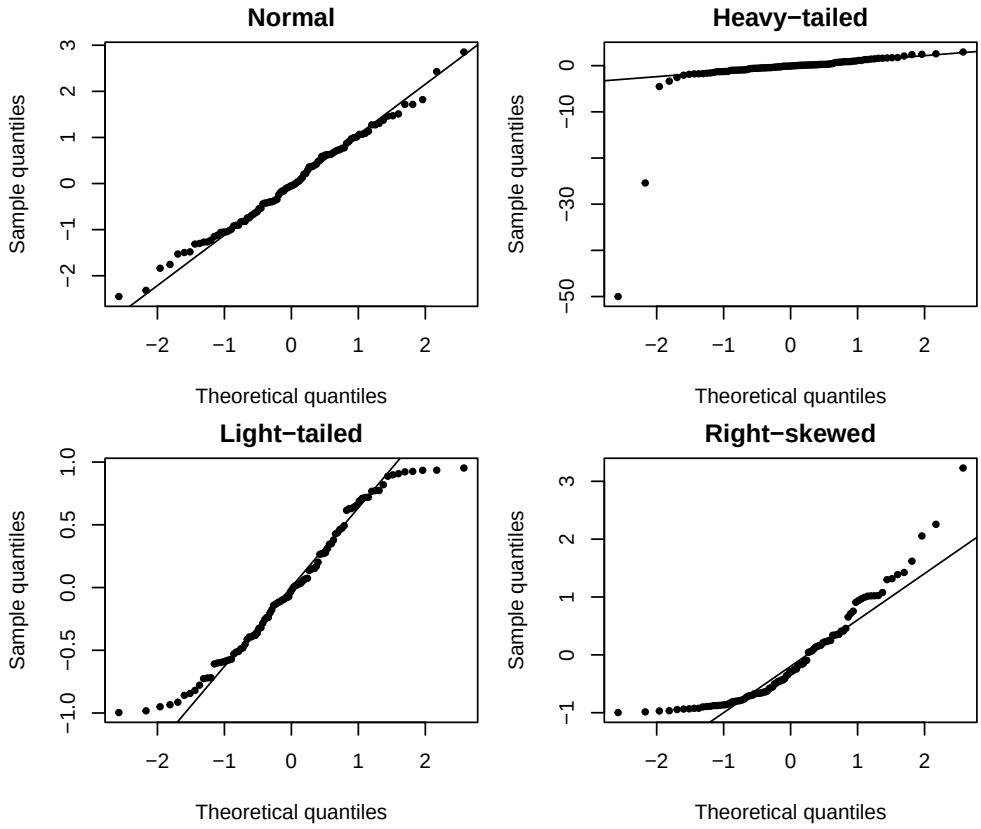
5.1 Normal Probability Plot

Normal Probability Plot: Construction

- Order the residuals: $e_{[1]} \le e_{[2]} \le ... \le e_{[n]}$
- Plot $e_{[i]}$ against the normal quantile $\Phi^{-1}\left(\dfrac{i - 1/2}{n}\right)$
- If the errors are normal, the points fall close to a straight line
- Use studentized or R-student residuals so that all residuals have the same variance

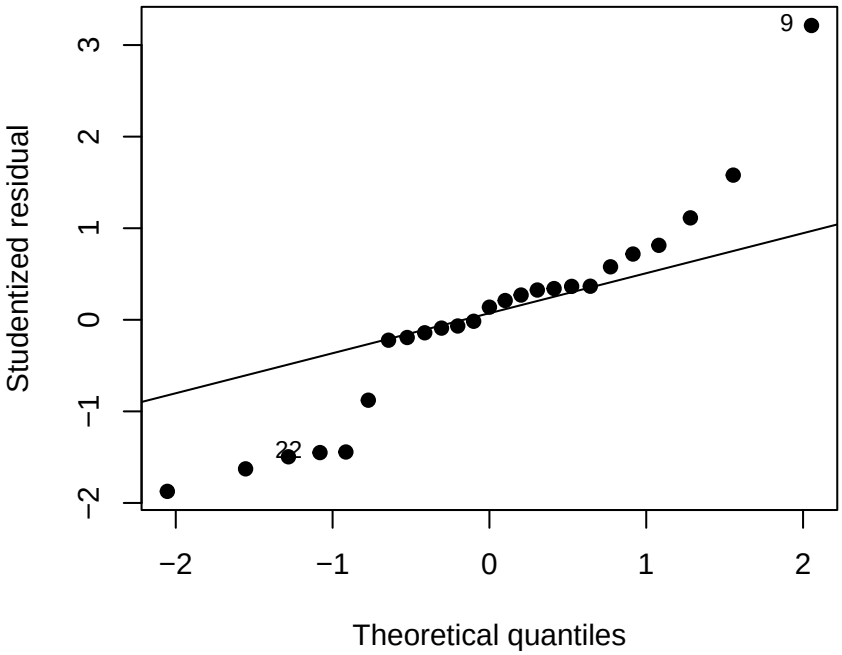
Normal Probability Plot: Patterns

- Straight line: consistent with normality
- Heavy tails: ends bend away from the line (S-shape)
- Light tails: ends bend toward the center (reversed S)
- Skewness: one-sided curvature
- Isolated points at the ends: outliers



Normal Probability Plot: Delivery Time Data

- The central points follow the line
- Observation 9 ($r_9 = 3.21$) lies far above the line in the upper tail
- Is the departure from normality serious, or is it one point?



5.2 Shapiro–Wilk Test

Shapiro–Wilk Test

H_0: the errors are normally distributed, against H_1: they are not.

$$W = \frac{\left(\sum_{i=1}^{n} a_i \, e_{[i]}\right)^2}{\sum_{i=1}^{n} (e_i - \bar{e})^2}$$

- $e_{[i]}$ are the ordered residuals; the constants a_i come from the expected order statistics of a normal sample and are supplied by software
- W is closely related to the squared correlation between ordered residuals and normal scores
- $0 < W \leq 1$: values near 1 are consistent with normality, small W rejects H_0

Test or Plot?

With $n = 100{,}000$ the Shapiro–Wilk test rejects for negligible departures from normality; with $n = 10$ it rarely rejects. Which should we trust, the test or the plot?

Delivery Time Data: Shapiro–Wilk

Applied to the studentized residuals: $W = 0.923$, $p = 0.060$.

- Do not reject normality at the 5% level, but the evidence is borderline
- The plot located the problem: observation 9
- Refitting without observation 9 (diagnostic only): $W = 0.970$, $p = 0.68$

6 Constant Variance

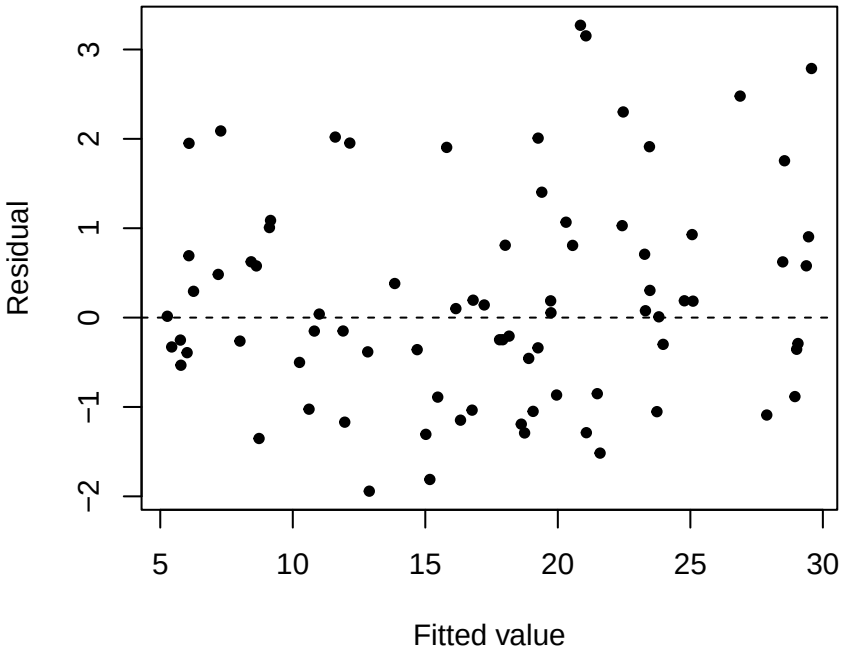
6.1 Residuals Against Fitted Values

Residuals Against Fitted Values

- Plot r_i (or e_i) against $\widehat{y}_i$
- If the model is adequate, the points form a horizontal band around zero with no structure
- Any systematic pattern points to a violated assumption

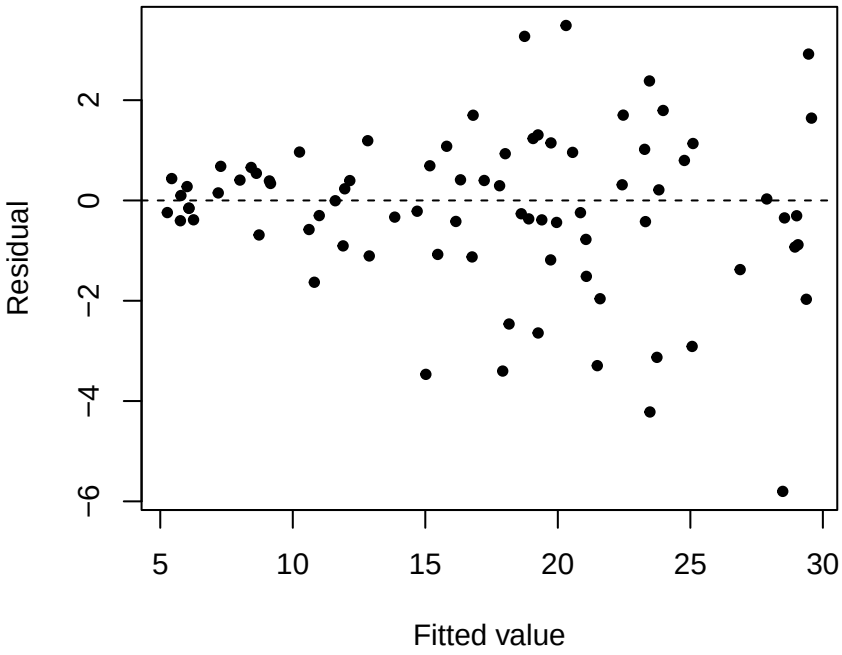
Pattern (a): Satisfactory

- Horizontal band centered at zero
- No trend, no change in spread
- No evidence against linearity or constant variance



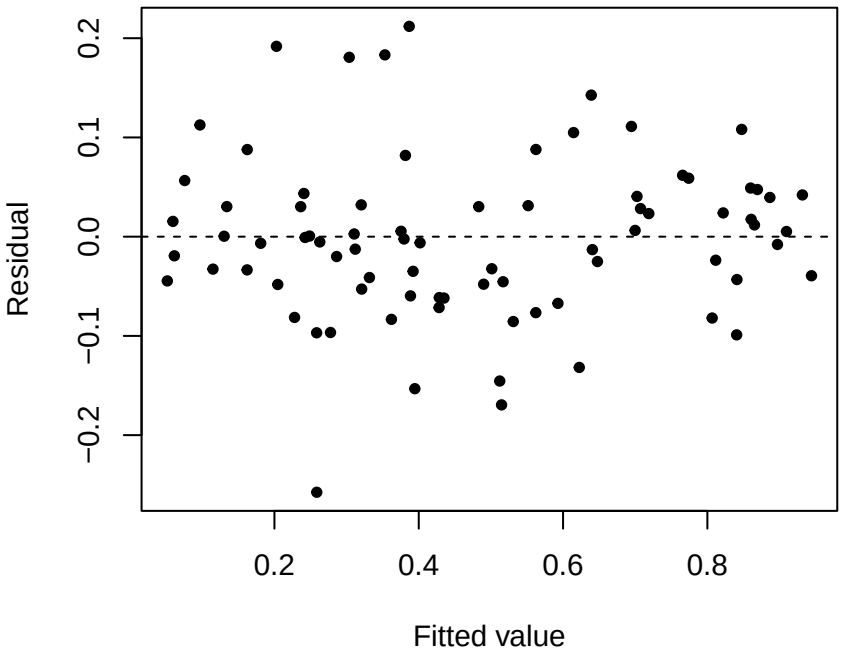
Pattern (b): Funnel

- Spread of the residuals grows with $\widehat{y}_i$
- The variance is not constant, typically $\mathrm{Var}(y)$ increases with $\mathrm{E}(y)$
- Remedies: variance-stabilizing transformation of y, weighted least squares



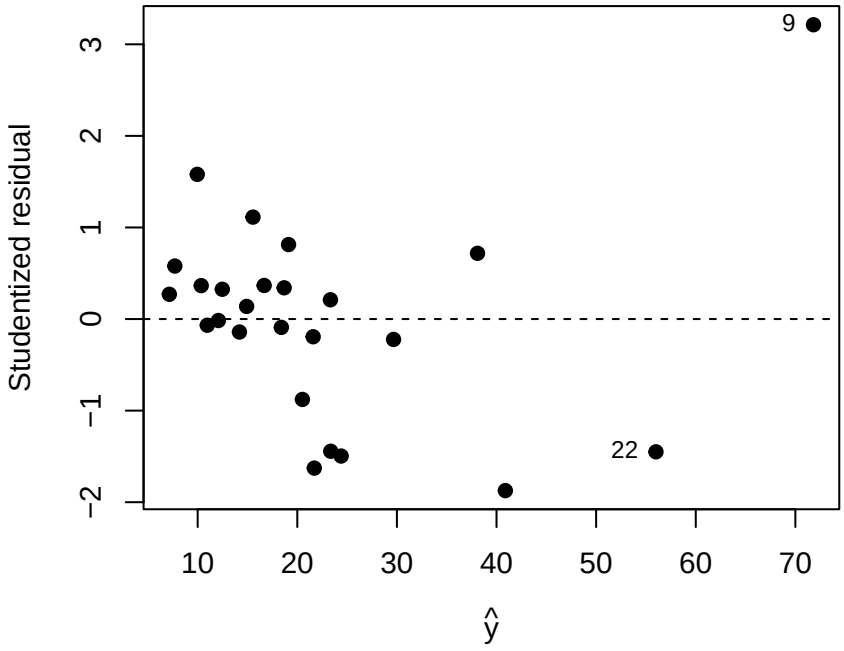
Pattern (c): Double Bow

- Spread is largest in the middle and smallest at both ends
- Typical when y is a proportion in $(0, 1)$, where $\text{Var}(y) \propto p(1-p)$
- Remedy: transformation of y



Delivery Time Data: Residuals Against $\hat{y}_i$

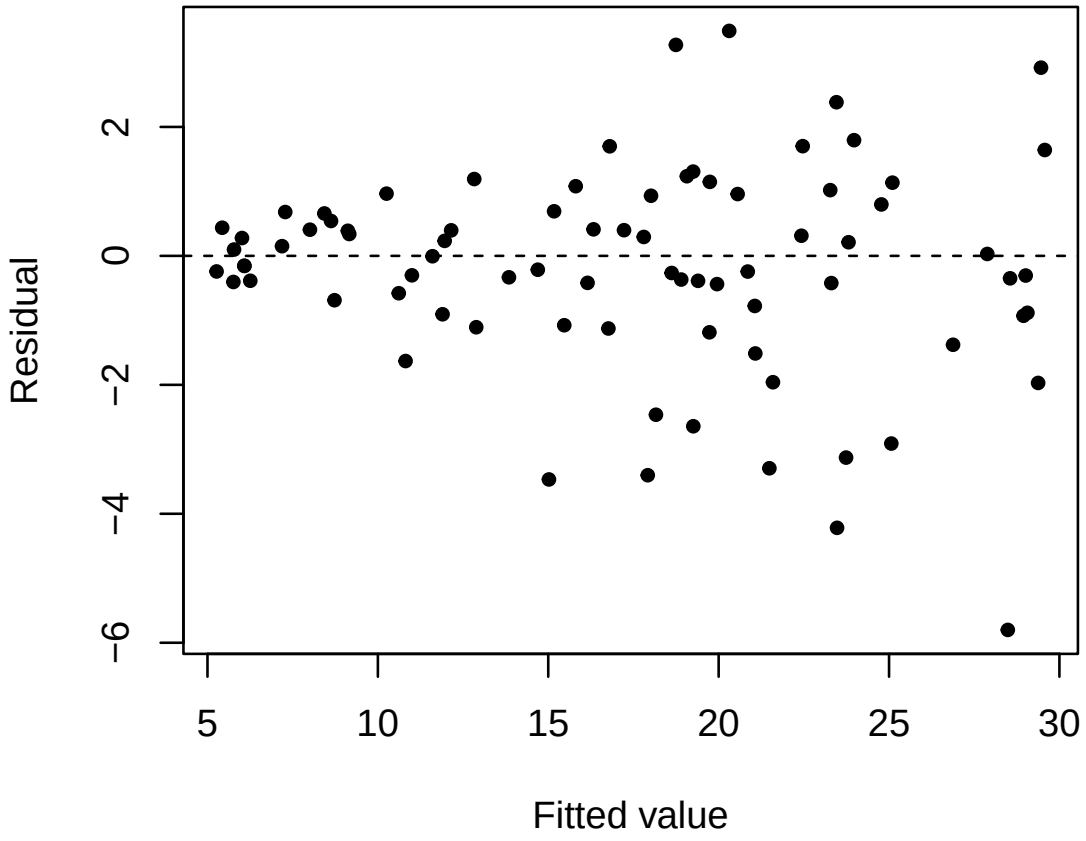
- Most points form a roughly horizontal band
- Observations 9 and 22 stand apart at large fitted values
- Is the spread at large $\hat{y}_i$ real, or driven by these two points?



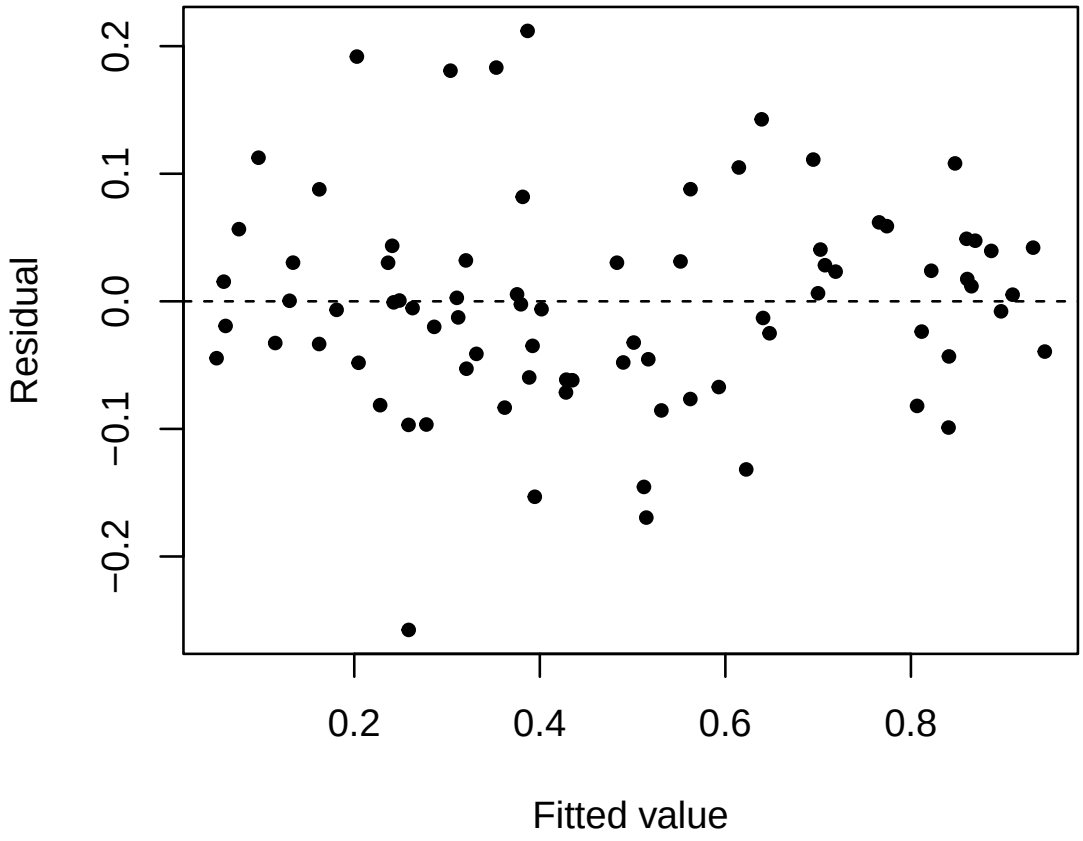
Name the Violation

Which assumption does each plot call into question?

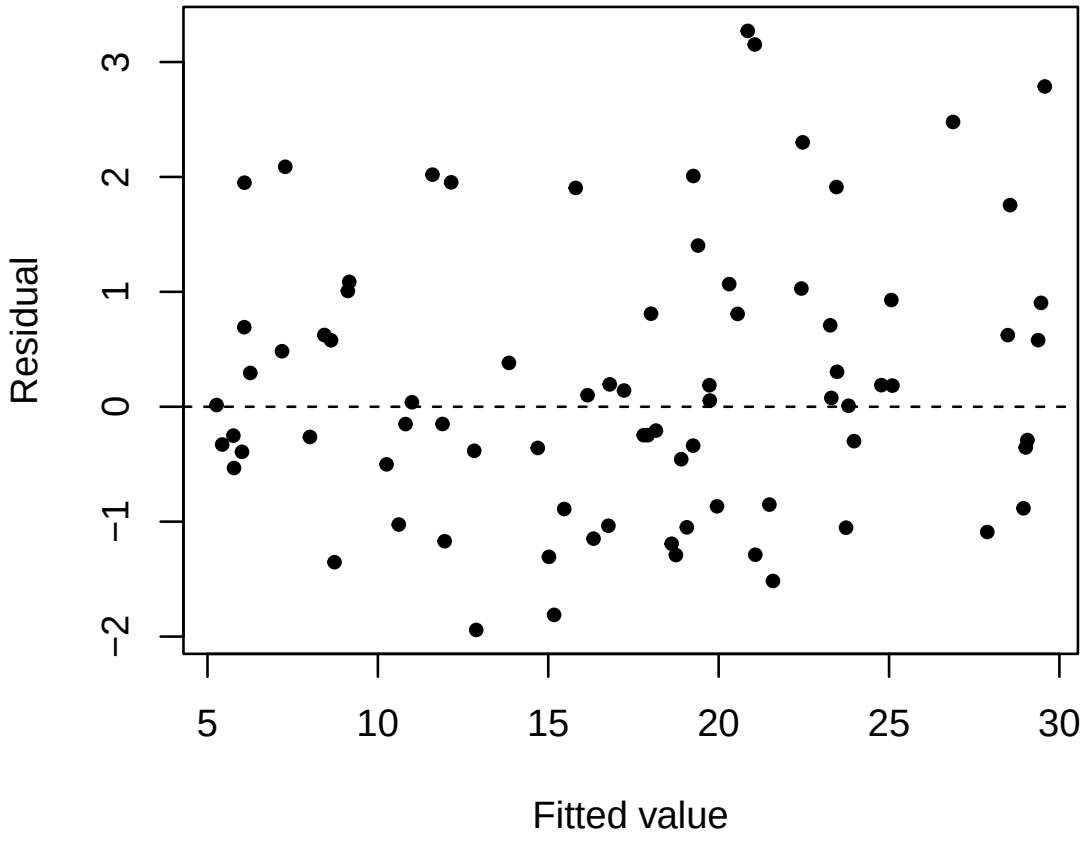
(i)



(ii)



(iii)



6.2 Breusch–Pagan and White Tests

Why a Test?

A funnel plot is a judgment call. Formal tests of

$$H_0 : \mathrm{Var}(\varepsilon_i) = \sigma^2 \text{ for all } i$$
$$H_1 : \text{the variance depends on the regressors}$$

Two common choices: Breusch–Pagan and White.

Breusch–Pagan Test

1. Fit the model and obtain the residuals e_i
2. Regress e_i^2 on the k regressors and record R^2_{aux}
3. Compute $LM = n\, R^2_{\text{aux}}$
4. Under H_0, $LM \sim \chi^2_k$ approximately for large n; reject for large LM

This is the studentized form, the default in `bptest` in R.

White Test

Same steps as Breusch–Pagan, but the auxiliary regression of e_i^2 uses the regressors, their squares and their cross-products. For $k = 2$ regressors

$$e_i^2 \text{ on } x_1,\ x_2,\ x_1^2,\ x_2^2,\ x_1x_2 \quad \Longrightarrow \quad LM \sim \chi^2_5.$$

- No assumption on the form of the heteroscedasticity
- Many auxiliary regressors when k is large, which lowers power

Delivery Time Data: Tests for Nonconstant Variance

Test	LM	df	p-value
Breusch–Pagan	11.99	2	0.0025
White	14.96	5	0.0105

- Both reject constant variance at the 5% level
- The rejection is driven by observation 9: refitting without it (diagnostic only) gives Breusch–Pagan $LM = 3.22$, $p = 0.20$

If the Variance Is Not Constant

- Variance-stabilizing transformation of y
- Weighted least squares (WLS)

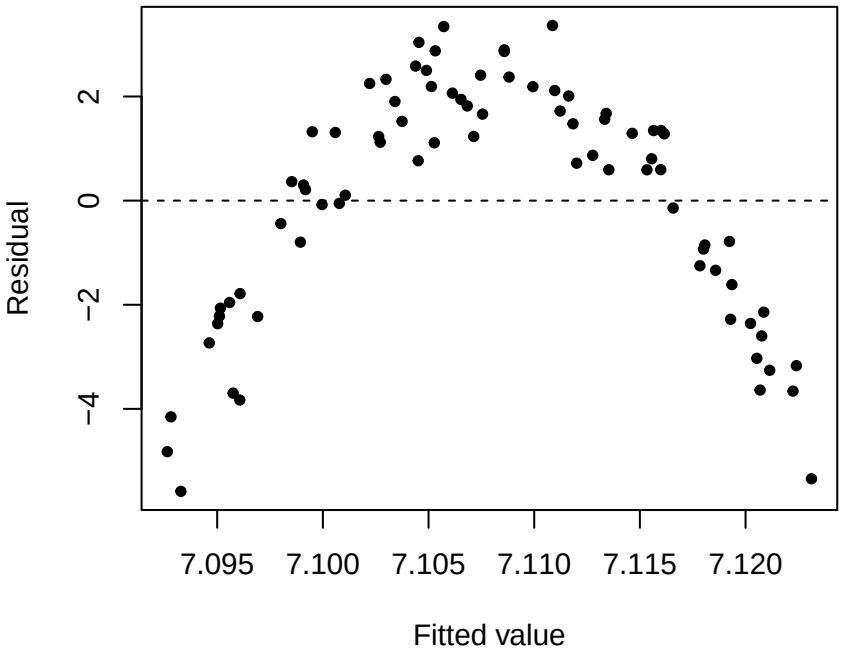
Before either, check whether a single unusual observation, as in the delivery time data, produced the rejection.

7 Linearity of the Mean Function

7.1 Residuals Against Fitted Values

Pattern (d): Curvature

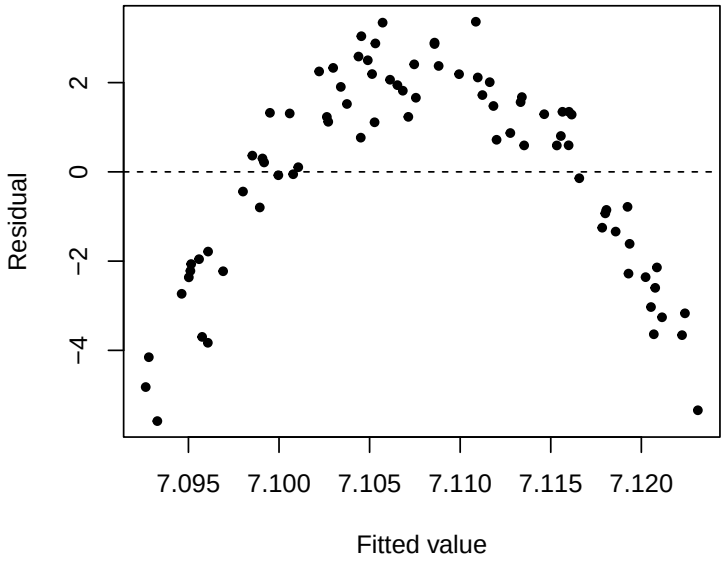
- Residuals are systematically positive in one range of $\hat{y}_i$ and negative in another
- The mean function is wrong: the linear model is inadequate
- Remedies: add higher-order or interaction terms, transform the regressors or y



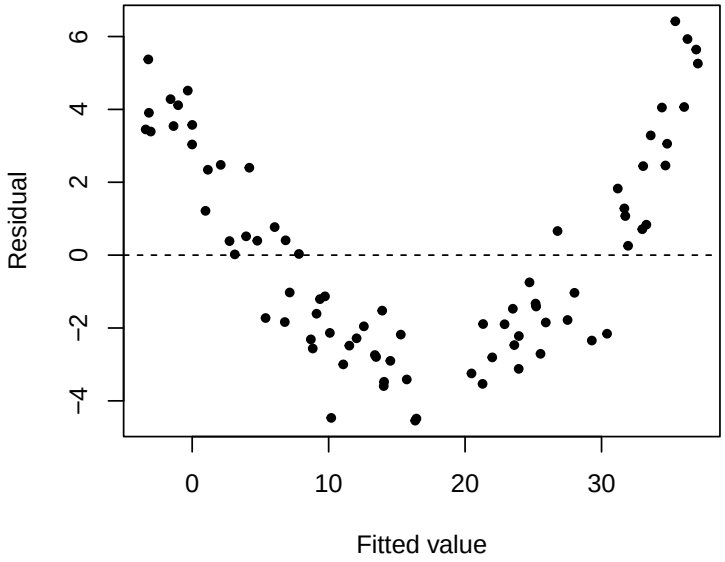
Name the Violation

Which assumption does each plot call into question?

(i)



(ii)



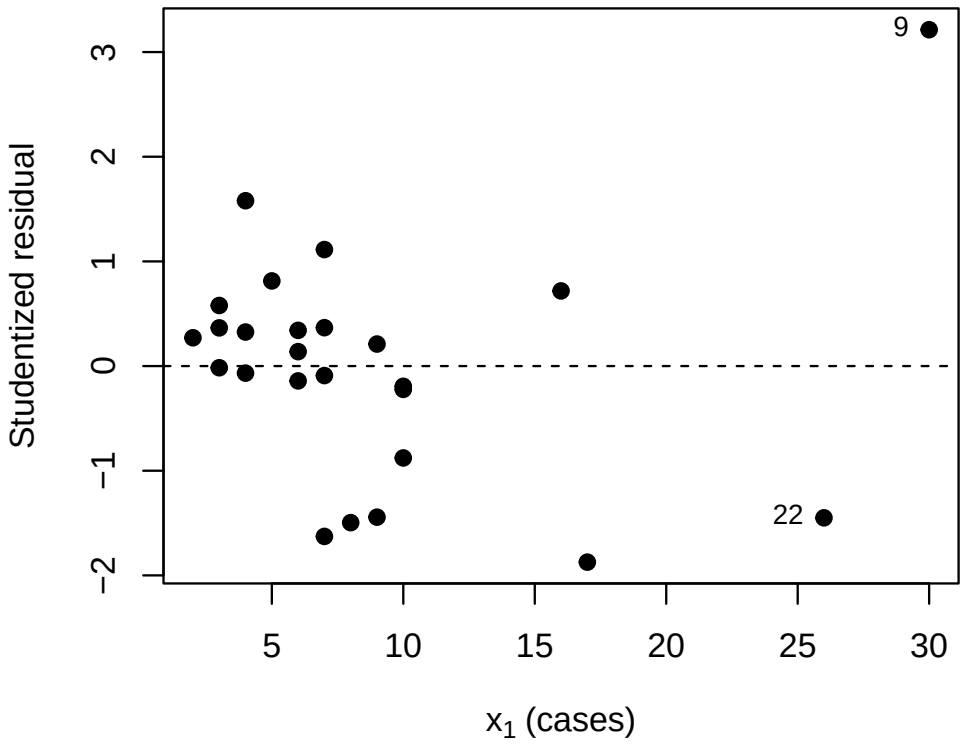
7.2 Residuals Against Regressors

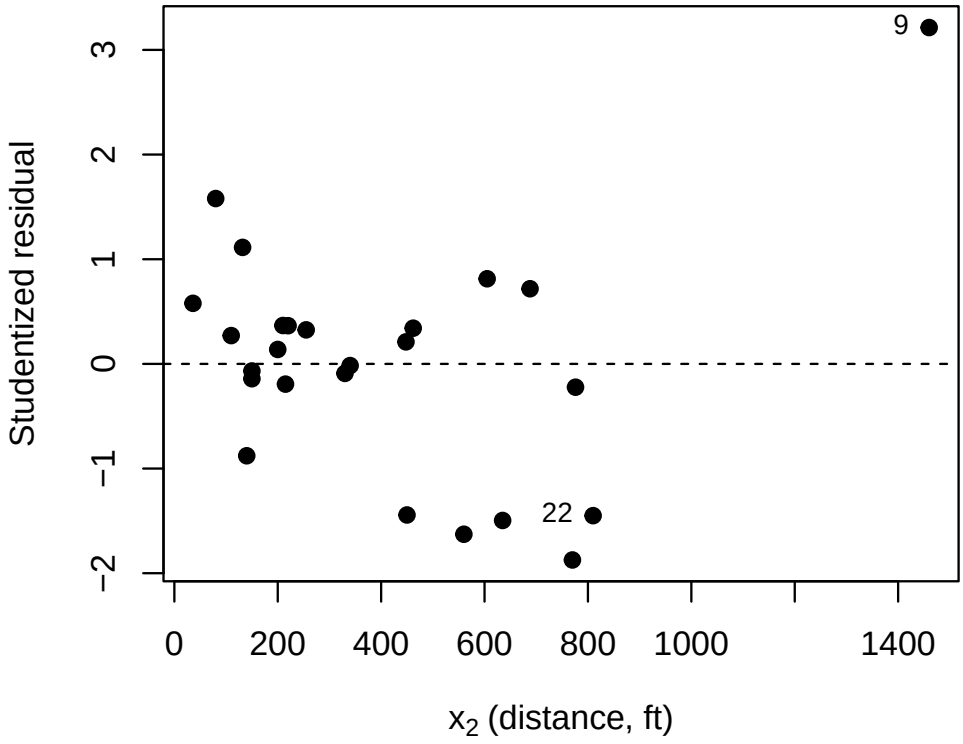
Residuals Against Regressors

Plot r_i against each regressor x_j.

- Curvature: x_j enters the mean function in a nonlinear way, add a higher-order term or transform x_j
- Funnel: the variance depends on x_j
- A plot against a regressor not in the model: a pattern means that regressor should be added

Delivery Time Data: Residuals Against x_1 and x_2





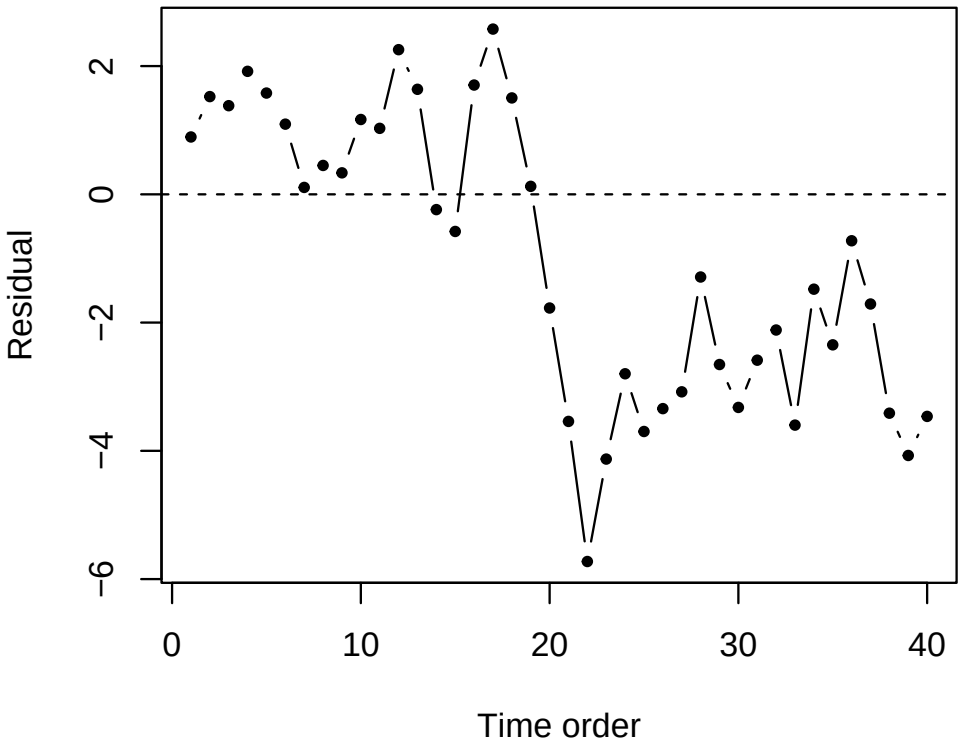
Neither plot shows clear curvature. Observation 9 is isolated at the extreme of both regressors.

8 Uncorrelated Errors

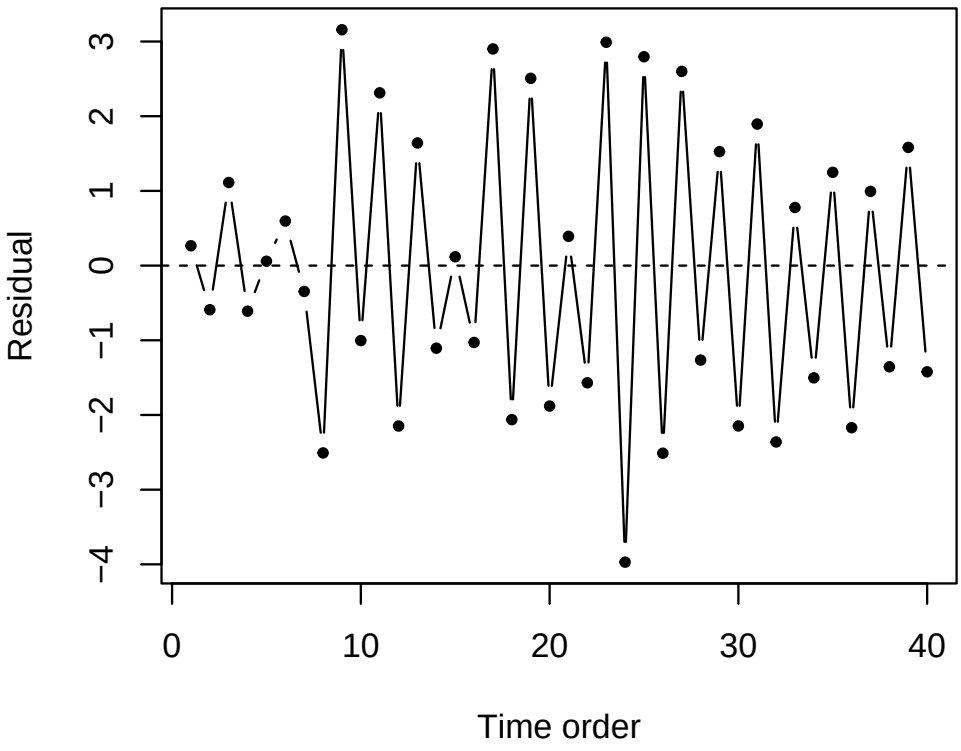
8.1 Residuals Against Time Order

Residuals Against Time Order

Plot e_t (or r_t) against the order in which the data were collected.



Positive autocorrelation: long runs of one sign.



Negative autocorrelation: signs alternate.

A trend over time suggests a missing time-dependent regressor.

8.2 Durbin–Watson Test

First-Order Autoregressive Errors

$$\varepsilon_t = \phi\,\varepsilon_{t-1} + a_t, \qquad |\phi| < 1, \qquad a_t \overset{iid}{\sim} N(0, \sigma_a^2)$$

$$\begin{aligned} \mathrm{E}[\varepsilon_t] &= 0 \\ \mathrm{Var}(\varepsilon_t) &= \frac{\sigma_a^2}{1-\phi^2} \\ \mathrm{Corr}(\varepsilon_t, \varepsilon_{t+j}) &= \phi^j \end{aligned}$$

$\phi > 0$: neighboring errors are similar (positive autocorrelation). $\phi < 0$: they tend to alternate in sign.

Consequences of Autocorrelated Errors

- The least-squares estimators are still unbiased, but no longer have minimum variance
- MSE typically underestimates σ^2
- The standard errors of the coefficients are underestimated
- Confidence intervals and t, F tests are no longer valid

Durbin–Watson Statistic

$$d = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \approx 2(1 - r_1), \qquad r_1 = \frac{\sum_{t=2}^{n} e_t e_{t-1}}{\sum_{t=1}^{n} e_t^2} \tag{5}$$

- r_1 is the lag-1 sample autocorrelation of the residuals
- $0 \le d \le 4$
- $d \approx 2$: no autocorrelation; $d < 2$: positive; $d > 2$: negative

Decision Rule

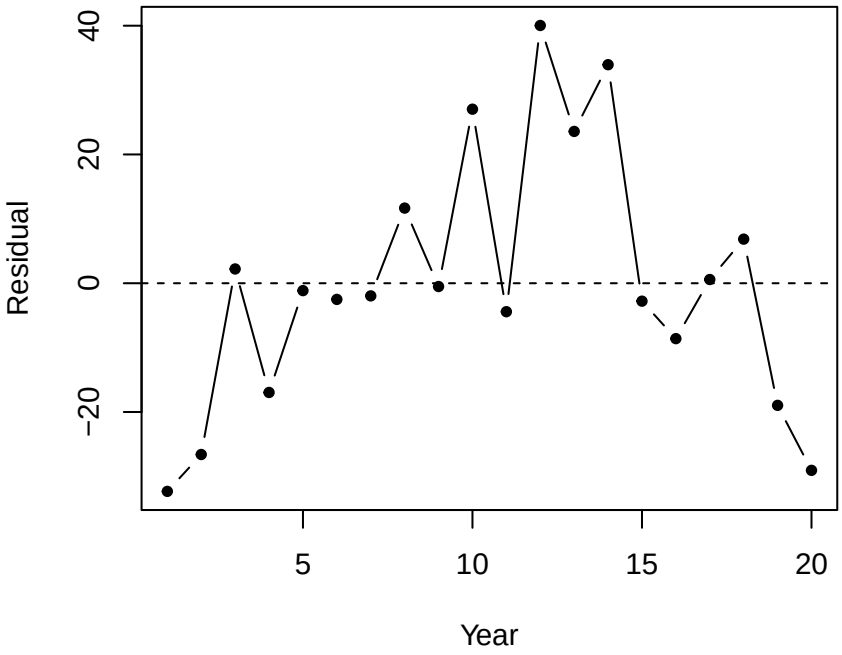
Test $H_0 : \phi = 0$ against $H_1 : \phi > 0$. The bounds d_L and d_U depend on n, the number of regressors k and α, and are tabulated.

- $d < d_L$: reject H_0, positive autocorrelation
- $d > d_U$: do not reject H_0
- $d_L \le d \le d_U$: inconclusive

For $H_1 : \phi < 0$ apply the same rule to $4 - d$, with d from (5).

Soft Drink Sales Data

- y: annual sales in a sales region, x: annual advertising expenditure, $n = 20$ years
- Fitted model: $\hat{y} = 1608.5 + 20.091\, x$
- $d = 1.08$
- $\alpha = 0.05$, $n = 20$, $k = 1$: $d_L = 1.20$, $d_U = 1.41$
- $d < d_L$: reject H_0, evidence of positive autocorrelation



The plot alone is hard to judge by eye.

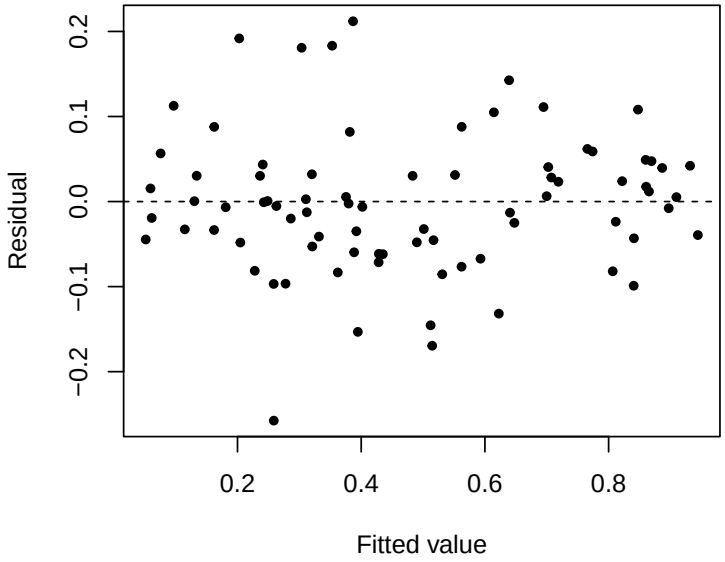
If the Errors Are Autocorrelated

- First look for a missing regressor, often a time-related one
- Otherwise estimate the model with an iterative procedure such as Cochrane–Orcutt (MPV, Chapter 14)

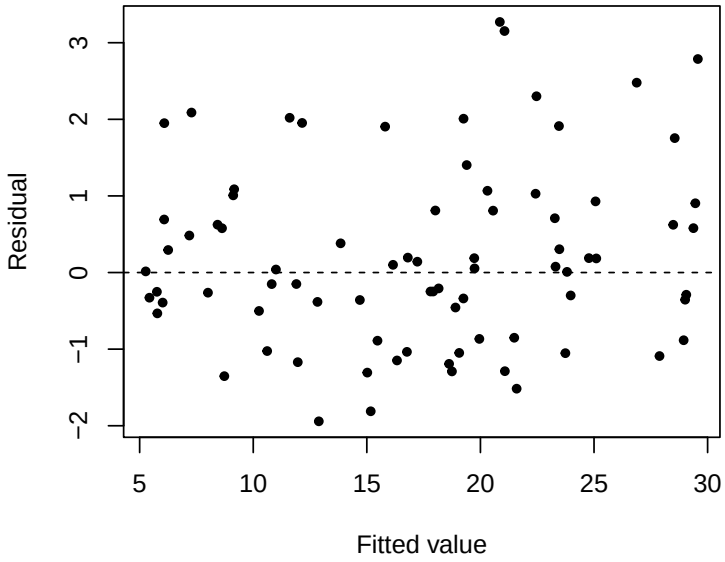
9 Summary

Review: Which Assumption Is Violated?

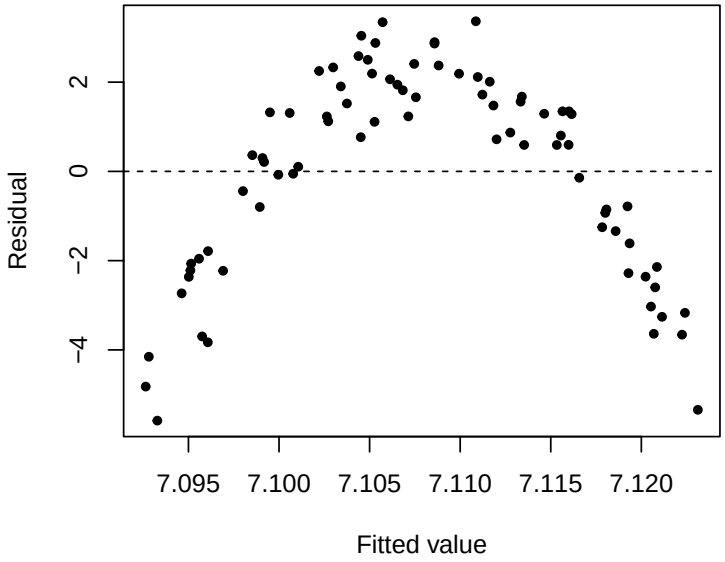
(i)


Residual
0.2
0.1
0.0
−0.1
−0.2
0.2
0.4
0.6
0.8
Fitted value

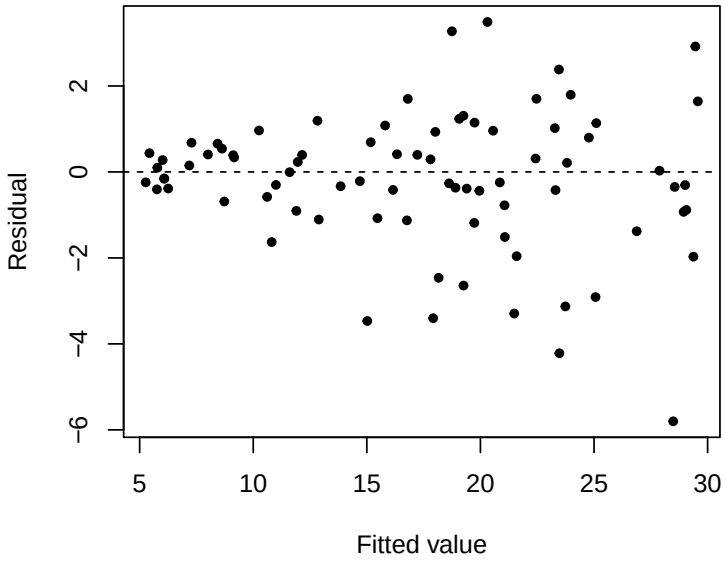
(iii)


Residual
3
2
1
0
−1
−2
5
10
15
20
25
30
Fitted value

(ii)



(iv)



Diagnostics at a Glance

Tool	Checks	Typical remedy
Normal plot, Shapiro–Wilk	Normal errors	Transform y
Plot vs $\hat{y}_i$, Breusch–Pagan, White	Constant variance	Transform y, WLS
Plot vs $\hat{y}_i$ and vs x_j	Mean function	Add terms, transform
Plot vs time, Durbin–Watson	Uncorrelated errors	Add regressor
r_i, t_i, $e_{(i)}$	Unusual observations	Investigate

Workflow and Key Messages

1. Fit the model and compute studentized and R-student residuals

2. Plot: normal probability plot, against $\widehat{y}_i$, against regressors, against time order
3. Use formal tests to back up what the plots suggest
4. Apply a remedy, refit, and check again

- Residuals are not the errors: they have unequal variances and are correlated
- Plots first, tests second; tests depend heavily on n